

DOCUMENT: User Requirements Specification for Equipment and Machinery**Tablet Coating Machines****PROJECT:** JV-Krka Pharma Pvt.Ltd.**Project Number:** 0706/2024**Project Location:** Hyderabad (Genome Valley – Hyderabad, India)**Project manager:** Zevnik, Gregor**Project manager Krka Pharma Pvt. Ltd.:** Perše, Boštjan**Date:** 15. 01. 2026**Prepared by (Authors):**

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1 PURPOSE

The purpose of this user requirement specification is to identify and document all the detailed requirements of Automatic Tablet Coating Machines as given by the user department. More precisely - the equipment used to coat the tablets with film coat.

2 SCOPE

The URS is applicable to Automatic Tablet Coating Machine which will be used to carry out coating of tablets, manufactured at JV-Krka Pharma Pvt.Ltd.

This document specifies the requirement for correct “Design, Manufacture, Install & Qualify” in line with cGMP.

Production area will be situated on the second floor (Level 3) with production rooms being ISO 8 class area and with installed equipment utilizing maximum level of protection of the operators and environment.

The spraying liquid is expected to have density in the range of 0,8 kg/L – 1,3 kg/L and viscosity up to 500 cP. However, several dry substances that are added to the water/solvent are considered difficult to mix-in and are mixed-in into a viscous liquid when the vessel is almost full. Therefore, special care must be taken regarding the effectiveness of the agitators for the mixing-in functionality (preparation vessels are not scope of supply).

Products to be produced in new facility are OSD products in form of granulates, capsules, tablets and film coated tablets. This document deals only with tablet coating phase.

The manufacturing program includes two different processes: organic solvent based, and water based.

- Organic solvent based: The spraying liquids are suspensions and solutions with or without active ingredients, organic solvents and additives. The solvent used in this process can be either ethanol (96%) or acetone or isopropanol.
- Water based: The products and processes are similar to the solvent-based technologies, but instead of organic solvents, water is used.

Explosion properties:

ETHANOL 96%	
Molecular formula	C ₂ H ₅ OH
Molar mass	46 g/mol
Relative vapour density (air)	1,59
Flash point	+12 °C
Explosion limits (LEL...UEL)	3.1 vol.%...14.3 vol.% 60 g/m ³ ...360 g/m ³
Min. ignition energy (MIE)	> 0.2 mJ
Vapour subdivision (equipment group)	IIB
Temperature class (min. ignition temperature)	T2 (400 °C)
Limiting oxygen concentration (LOC)	9.3 %
Max. explosion pressure (p _{max})	8.4 bar

ETHANOL 96%

Electrical (volume) conductivity	0.135 µS/m
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ISOPROPANOL

Molecular formula	C ₃ H ₇ OH
Molar mass	60 g/mol
Relative vapour density (air)	2,07
Flash point	+12 °C
Explosion limits (LEL...UEL)	2 vol.%...13.4 vol.% 50 g/m ³ ...335 g/m ³
Min. ignition energy (MIE)	0.49 mJ
Vapour subdivision (equipment group)	IIA
Temperature class (min. ignition temperature)	T2 (425 °C)
Limiting oxygen conc. (LOC)	8.7 %
Max. explosion pressure (p _{max})	7.8 bar
Electrical (volume) conductivity	5.8 µS/m

ACETONE

Molecular formula	CH ₃ -CO-CH ₃
Molar mass	58 g/mol
Relative vapour density (air)	2,00
Flash point	-20 °C
Explosion limits (LEL...UEL)	2.5 vol.%...14.3 vol.% 60 g/m ³ ...345 g/m ³
Min. ignition energy (MIE)	0.55 mJ
Vapour subdivision (equipment group)	IIA
Temperature class (min. ignition temperature)	T1 (535 °C)
Limiting oxygen conc. (LOC)	
Max. explosion pressure (p _{max})	9.7 bar
Electrical (volume) conductivity	0.49 µS/m

METHANOL

Molecular formula	CH ₃ OH
Molar mass	32 g/mol
Relative vapour density (air)	1,10
Flash point	+9 °C
Explosion limits (LEL...UEL)	6 vol.%...50 vol.% 80 g/m ³ ...665 g/m ³
Min. ignition energy (MIE)	0.14 mJ
Vapour subdivision (equipment group)	IIA
Temperature class (min. ignition temperature)	T2 (440 °C)
Limiting oxygen conc. (LOC)	8.1 %
Max. explosion pressure (p _{max})	8.5 bar
Electrical (volume) conductivity	0.15 µS/m (conductive)

3 PROCESS / PRODUCT / SYSTEM

3.1 Name of Equipment

Nr.	Equipment designation	Name of Equipment	Room	Remark
1	Automatic Tablet Coating Machine - Pilot	Pilot Tablet Coating Machine	Coating - 1	Batch Coater with interchangeable drums
2	Automatic Tablet Coating Machine - 1	Tablet Coating Machine - 1	Coating - 2	Batch Coater, WIP system
3	Automatic Tablet Coating Machine - 2	Tablet Coating Machine - 2	Coating - 3	Batch Coater, WIP system

The scope of these URS and scope of supply must include:

1. The above stated units.
2. All connections points/interfaces preparation, including counter flanges to utilities, but without materials and without execution of connection (erection of equipment and execution of connections is in Krka Pharma's scope of supply)
3. Supervision and engineering support for erection and installation and connections must be included in quotation.
4. An obligatory part of the quotation is at least rough estimation of the media supply needed by the machinery (media type, average consumption, peak consumption) and needed space for installation and operation of the equipment (main dimensions of the equipment, including weight of the equipment and clearance around the equipment, necessary for normal operation).
5. The attached layout of the room available is showing one of the possible versions of room arrangement. In order to check the arrangement and to maybe optimize the positioning of the equipment in the room available, an AutoCAD floor print/layout of the equipment is required. Without this information the quotation cannot be processed.
6. The quotation must clearly show value of each individual position and end sum value as well. Quotations with just lump sum will not be processed.

3.2 Capacity

Nr.	Equipment designation	Working volume
1	Automatic Tablet Coating Machine - Pilot	10 L – 60 L
2	Automatic Tablet Coating Machine - 1	100 L - 500 L
3	Automatic Tablet Coating Machine - 2	300 L - 1000 L

3.3 Purpose of equipment

This machine provides coating of the tablets in general OSD production plant.

3.4 Production concepts

3.4.1 Generally

The equipment delivered must offer maximum possible protection for the operators and environment, to enable operators to work in the facility with minimum personal protection equipment.

The production of the products is expected to be limited to batch sizes from appr. 100 - 800 kilograms. Therefore, all material transports are expected to be manual, and the material flow is set up as "horizontal" type of production.

Transport of materials will be done in IBCs (for 1000 L tablet coater only) and/or in PE/ALU bags.

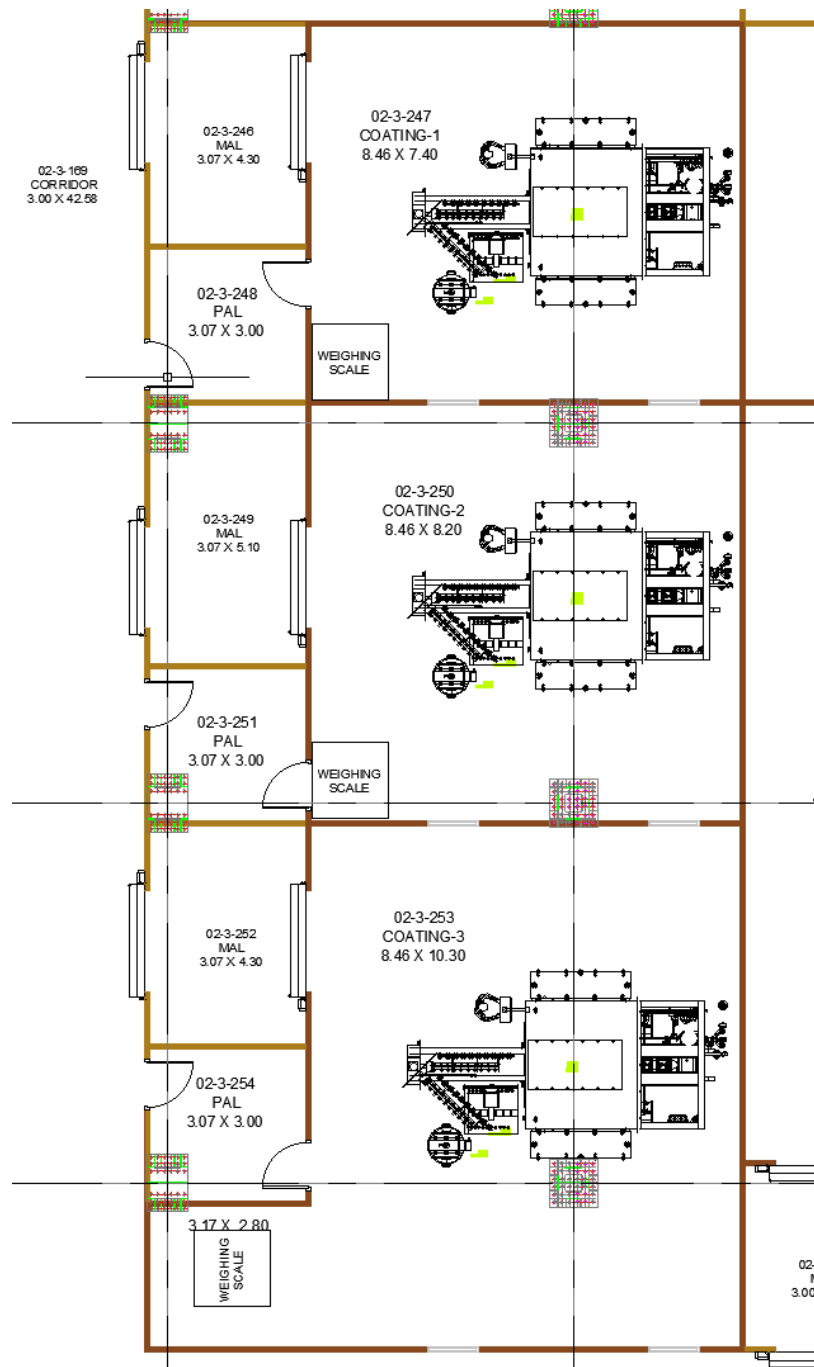
3.4.2 Film coating of tablets

1. The coating liquid (suspension or solution) will be prepared in a dedicated mixing vessel (not in scope of supply of coating machine).
2. Tablets are transferred into the pan of coating machine manually (Pilot and 500 L machine) or through the IBC.
3. Coating liquid is supplied under pressure (adjustable) to the spray-dosing system (presumably peristaltic pumps) and is sprayed to the bed of tablets. Tablets are simultaneously dried with hot air blowing through the pan.
4. After coating is finished, the pan should be automatically emptied through the discharging nozzle to the receiving PE/ALU bags.

The process side of machine is located in clean area (class D).

3.4.3 Available Coating areas

Coating area is constructed as D class area (cleanliness class ISO 8) area. Other rooms are connected by a corridor with overpressure regarding to the coating machine rooms. Rooms available for coating machine are is roughly 8,5m x 7,5 (up to 10m) m (height appr. 4,0 m).

**Figure 1 Available rooms for tablet coating machines**

3.5 Process/Equipment Requirement

3.5.1 General

1. Coating Machine is considered to be a "system", consisting of:
 - Air Handling Units (Supply and Exhaust).
 - Coating pan with spraying equipment.
 - Outfeed air treatment (deduster with filter).

- Scrubber.
 - WIP system.
2. All the parts of the production equipment that come in contact with product or with input materials should be made of AISI316L (EN 1.4404 or EN 1.4435) or better. All other parts should be made of AISI304 (EN 1.4301) if applicable. The surface of product wetted parts should be electro polished ($Ra < 0,8\mu m$) and passivated. The non-product contact parts and outside of the vessels should have $Ra < 1\mu m$ including the welds and seams in material which should be polished.
 3. Non-metal parts in contact with the product (such as flexible hoses and sealing) should be made of silicone or Teflon. The silicone flexible hoses in contact with the product should be made of platinum cured silicone and all gaskets (for Tri-clamp connections) should be PTFE covered EPDM.
 4. Parts being in contact with product as mentioned in the paragraphs above are considered to be all pipes, valves, connections,... in contact with purified water, solvents or any ingredient or coating liquid.
 5. For the connection of the pipes smaller than DN80 a standard DIN tri-clamp type of connection is preferred.
 6. The spray tubes shall be made of food grade silicone rubber and shall be withstand the solution pressure on the wall of tubes.
 7. Spray guns shall be mounted on header assembly with return line to solution feed vessels with silicone tubing.
 8. The pressure tube connection for compressed air and spray solution shall be provided with manifold, which shall be manually operated for removal and insertion. The gun bar shall be enclosed in the main cabinet which shall be detachable for cleaning.
 9. All edges of the equipment shall be rounded at corners.
 10. All pipe joints shall be sanitary with tri-clover fitting.
 11. The air handling shall be in such way that negative pressure maintained inside of the pan.
 12. Auto On/Off Drain shall be provided the scrubber tank's bottom with suitable valve arrangement.

3.6 Cleaning of the Machines

The Tablet Coating Machines should be equipped with a WIP system (500 L and 1000 L machines only) with an option to connect two detergents. This means, that installed WIP system must be able to wash all the major parts of the tablet coating machine.

- The quotation should specify the expected needed media.
- The quote should specify the expected battery limits and media needed (with consumptions parameters) to execute the cleaning of the units.
- Washing should be performed in automatic way by splashing of cold purified and hot process water over spray nozzles installed and with detergent.

- Design of the equipment inside the contaminated area must allow efficient cleaning which mean that:
 - a. No dead spots or corners should be present.
 - b. All surfaces must be accessible for manual cleaning.
- WIP system station must be offered.

3.7 cGMP Requirements

1. Should be operate in closed condition to avoid contamination and cross contamination of the product(s).
2. Manufacturer's name, address, major specifications, equipment serial no. and model no, whichever is applicable should prominently appear on the equipment.
3. Equipment shall be easy to dismantle and clean.
4. Calibration certificate for all instruments which are located on the equipment's.
5. Data backup, restoration, security and privileges shall be complies with 21 CFR Part 11 Compliance.

S. No.	Description of Component	Features / MOC / Surface finish
1	Product contact parts	SS316/ food grade plastic/ PU Where ever not possible supplier shall use corrosion resistant, strong alloy metals which are universally acceptable and certified along with food grade certificates.
2	Non-contact parts	All non-contact part should be of SS 304.

3.8 SW/HW Requirements

S.No	Component / Features	Description (As applicable)
1	Control Panel	Appropriate control panel should be provided along with the machine.
2	Protection class	Is acceptable as per manufacturer's standard.
3	Indicator(s)	As per manufacturers standards.
4	Alarms and Warnings	All Critical and non-Critical alarms to be listed with proper description and solution.
5	IPC/MMI / HMI provision	8 GB and above RAM,1TB hard disc with 17 processor should be provided. preferable HMI make (Pepper+Fuchs or Siemens) with Keyboard and resistive touch mouse pad.
6	Operating system	Windows 11 should be provided with Licence Document.

7	Database	Specify the data storage Database with Licence
8	PLC	Preferable PLC make's (Siemens, Allen Bradley, Mitsubishi).
9	Interface to other system (s) / equipment (s) / instrument (s)	System PLC should have the provision of open Configurable TCP/IP (Transmission control protocol/Internet protocol).
10	Frequency drives (VFD'S)	Preferable VFD make's (Siemens, Schneider, Mitsubishi).
11	User interface - compatible to SCADA system.	Suitable interfaces for acquisition (Integration with Open Platform Communication system).
12	Training	Training should be provided at the time of Qualification.
13	MES system	MES system integration to be provided as per KRKA Pharma requirement

3.9 Safety requirements

S.No.	Description of Component	Features
1	Emergency Stop	As per industrial standards.
2	Power Failure/ Recovery	System shall be provide the total set values in loaded recipe on power resumption.
3	Electrical Protection (Flame Proof)	As per Manufacture standard.
4	Earthing	As per electrical standards.
5	Noise level	Less than 90 dB
6	Interlocks	Interlocks shall be provided.
7	Others	Protective guards shall be provided for all rotating parts.

3.10 21 CFR Part 11 and EU Annex 11 compliance requirements

Req. No.	Specifications
GMP1.00	User Interface
GMP1.01	The steps followed during the application development life cycle and integration with the Equipment / instrument shall be provided.
GMP1.02	Validation documents and reports available during the application development life cycle shall be provided.
GMP1.03	Risk associated with the application and mitigation plans are being carried out to meet the safety, data integrity and quality. Assessment document shall be provided.

GMP1.04	Application should be configurable OR bespoke /custom development for Krka Pharma.
GMP1.05	Post go-live (application) support shall be provided.
GMP1.06	SLAs and roles and responsibilities to support the system shall be provided during the implementation and post implementations of the system.
GMP1.07	Other than application relevant Area of HMI / CPU not accessible to users, except for Administrator.
GMP1.08	Ensure that, system requires sequenced steps, it ensures that the actions are performed in the correct sequence.
GMP1.09	Provide justification and documentary evidence of the system GAMP5 category. (1, 3, 4, 5)
GMP1.10	The system type to be mentioned Open-ended / closed.
GMP1.11	Login methods to be mentioned. Example – User ID & password or User's Biometric impressions or Access Cards / Tokens
GMP1.12	The SCADA / PLC, should allow to store the electronic data to a central server, in-parallel to the local computer system. (Real-time data backup).
GMP2.00	User Accounts
GMP2.01	System access should be limited to authorised individuals with User specific Application level - Login ID and Password.
GMP2.02	The system administrator shall have the ability to create and manager user Accounts and groups.
GMP2.03	A user account may include the alphabet/numeric of the user.
GMP2.04	The system administrator shall be able to reset user account Passwords.
GMP2.05	Each user account must be assigned privileges based on a specific user type, as defined in the procedure
GMP2.06	Application / System should enforce to change the password by the user – at the time of first login OR when Administrator resets the password.
GMP2.07	The system shall confirm that all users have a unique username/account ID (User ID).
GMP2.08	As per Electronic Record/Electronic Signature compliance requirements, system policies should be configurable to enforce: 1. Expiry date set to 'n' days (Password age) 2. Failure Login attempts/tries to 'n' times 3. Minimum Password length is 'n' Characters. Most recently used 'n' passwords should not be allowed. (Password history)
GMP3.00	User Privileges
GMP3.01	Software access can be controlled by user group.
GMP3.02	Access control must limit the level of access permitted to each user with regard to use of the software with their user group specified privileges.
GMP4.00	Audit Trail
GMP4.01	Audit trails must capture events that affect the data validity (for example- unauthorized users trying to access the system).
GMP4.02	The audit trail is required to record changes made and to request a reason statement from the person making the change.
GMP4.03	Audit trail entries are required for changes made to Particle size systems and acquisition systems
GMP4.04	Audit trail entries are required for changes made to users, user types, or user groups.

GMP4.05	Audit trail entries are required for changes made to methods and results
GMP4.06	Audit trail comment containing only spaces or blanks are not permissible
GMP5.00	Data Integrity
GMP5.01	It must be possible to Backup and Restore all data includes Audit trail.
GMP5.02	It must be possible to view the sample list and select the suitable template.
GMP5.03	Each data entry must be supplied with a time/date stamp. All data sets must reflect the identity of the responsible user.
GMP6.00	21CFR Part11 requirements
GMP6.01	Ability to generate records in human readable and electronic formats
GMP6.02	The system shall allow users to login using unique identity and password combination.
GMP6.03	The system must allow system administrators to specify a request for password change during initial login.
GMP6.04	The system must lock user accounts after specific incorrect login attempts.
GMP6.05	The system should provide capability for the configuration of different security access levels for the different type of users that may be configured on the system.
GMP6.06	The system shall force the user to change the password periodically.
GMP6.07	Audit trails can capable to record date and time of operator, entries and actions that create, modify or delete electronic records.
GMP7.00	Data Backup & Archival
GMP7.01	System should have ability to connect with Krka Pharma network and assign Krka Pharma IP address, to support auto backup to Krka Pharma backup server / storage
GMP7.02	User should not able to save or relocate the data files, into another location and it should be controlled through software only.
GMP7.03	Backup facility - The data should be backed up to a central network server using Automatic tools.
GMP7.04	System should be provided with automatic back up / process to back the data in to back up server or allow the Krka Pharma to install the automatic backup tool / agent?
GMP7.05	Recovery - Full data restoration/recovery procedures from held backup data should be available.
GMP7.06	Disaster Recovery: Allow the recovery of the system from a periodic system backup.
GMP7.07	Application should have data archival procedure (to archive old data.
GMP7.08	Application should have data restoration procedure (to restore archived data.
GMP7.09	Archived data should consist of Audit trails associated with the corresponding electronic record.

Note: Documented evidence to be provided to comply the above points wherever it is applicable.

3.11 Documentation requirements

S.No	Document details	Required (P / R)
1	Design Specification	⊗
2	Functional Specification	⊗
3	Qualification Documents (IQ,OQ)	⊗
4	PLC Alarm/Interlock/Safety/ communication/power failure test procedures	⊗
5	Process and Instrumentation Diagram (P&ID)	⊗
6	Instrument Listing	⊗
7	Electrical Panel Assembly Drawings	⊗
8	Machine area layout Drawings	⊗
9	Operator, Maintenance and Service Manuals	⊗
10	Spare Parts List	⊗
11	MOC certificates	⊗
12	Calibration certificates of instruments	⊗
13	Test certificates of components/test devices	⊗
14	Surface Area Calculation sheet of product contact parts	⊗
15	Latex and TSE / BSE certificates	⊗
16	GA drawing	⊗
17	Isometric drawing Optional	⊗
18	Electrical drawings	⊗

4 OPERATIONAL / DESIGN REQUIREMENTS

4.1 Tablet Coating Machine

4.1.1 General

1. All Process shall operate as per set parameters in recipe from IPC.
2. Operation of the machine shall be vibration free.

4.1.2 Cabinet

1. SS304 angular structure of suitable size, SS304 sheet with a smooth mirror polished surface inside and smooth dull polished surface outside.
2. The corner of the cabinet both inside and outside shall be rounded.
3. All the doors shall be pressed for proper sitting and fixing without hinges projecting outside.

4.1.3 Pan

1. SS316 baffles shall be provided for proper mixing suitable with 20% to 100% occupancy with different shape of tablets and turnover of tablets.
2. During pan rotation tablets may turnover. So, all edges of the baffles shall be smooth.
3. Rotating pan shall be provided with suitable static discharge device.
4. Gun arm shall be rotatable at header assembly for easy cleaning and maintenance.

4.1.4 Spray Guns

1. Suitable gun (Schlick make is preferred) with air nozzle (Gun orifice shall be 1.2mm, 1.5mm, 1.8 mm), SS316 L shall be provided.
2. Spray nozzles to be anti-bearded and easy clean.

4.1.5 Pan Drive

1. Pan drive shall be driven by sufficient capacity flame proof induction motor from reputed manufacturer, gear box and AC frequency converter with infinite pan speed control.

4.1.6 Pan Light

1. Pan light shall be provided with flame proof for proper illumination in the pan to view the spray and coating quality.

4.1.7 Pressure Differential

1. Adequate differential pressure shall be maintained accordingly.
2. Differential pressure gauges (Magnehelic gauge) shall be provided for the indication of the DP across the Pre & Fine filters.
3. Differential pressure transmitter shall be provided for the DP across the HEPA filter.

4.1.8 Spray unit

1. Spray unit shall be provided with pressure regulators (for all the spray guns separately).
 - a. Each dosing head (spray gun) on the peristaltic pump has to be pressure controlled with the HMI visualisation, enabling operator to see if clogging of the spray gun has occurred. Possibility to set up clogging range.
2. Retractable arm manually operating and retractable arm with angle adjustment for spray header shall be provided
3. Peristaltic pump shall be mounted on separate trolley with castor wheels.
4. Multi head peristaltic pump for coating liquid, including mass-flow meter to control the spraying rate, should be installed. A Coriolis mass flow meter is expected – E+H Promass P300 family of flow meters is preferred. The flow of the coating liquid must be controlled by the speed of peristaltic pumps. Expected flow rate of coating liquid is between 200 g/min and 1500 g/min (for

500 L machine), 300 g/min and 2500 g/min (for 1000 L machine) and 10 – 500 g/min (60 L machine).

5. Distance and angle of spray guns have to be controlled, with automatic set-up correction allowed (for 500 L and 1000 L line).
6. Two sets of tubes for coating liquid transfer must be offered. The final dimensions (diameter, wall thickness) will be defined during the DQ.

4.1.9 Inlet Air Handling Unit

1. Inlet air handling unit consists of inlet blower, De-humidification coil, Steam coil, Face & Bypass damper.
2. Dry air at 3 g of H₂O/kg of air should be achieved.
3. Inlet air between 20°C and 90°C is required.
4. At the end of the coating process, a cooling of the tablets should be allowed for. Expected temperature of the cooling air should be as low as possible. Typically, we reach this temperature as low as 20°C using chilled water with 7°C.

4.1.9.1 Inlet Blower Motor

- Blower fan impeller shall be dynamically & statically balanced.

4.1.9.2 Steam Coil

- Stainless steel steam coil shall be provided with pneumatic actuator for raising temperature from ambient (20°C to 90°C) and also steam trap shall be provided.

4.1.9.3 Filters

- 3 stages filtration shall be provided (Pre Filter, Fine Filter, and HEPA filter) for purification of incoming fresh air. The following filter specifications are given below.
 - a. Pre-filter up to 10 micron with differential pressure gauge.
 - b. Intermediate filter or fine filter up to 5 micron with differential pressure gauge.
 - c. HEPA filter 0.3μ, 99.97% efficiency with differential pressure transmitter.
 - d. Provision for PAO test shall be provided.
 - e. The filters shall be withstand 110% of the blower capacities.
 - f. Light shall be provided to inside of the inlet blower section.

4.1.9.4 Ducts:

- SS 304 insulated rounded ducts from inlet air unit to the pan and from pan to exhaust air unit shall be provided.
- SS damper with pneumatic actuator ON/OFF control shall be provided.
- All ducts shall be made of more than 2.5mm thick sheet.

4.1.10 Exhaust Air Unit

- Exhaust Blower motor

- Capacity: Suitable CFM blower shall be provided.
- Type: Flame proof shall be provided.

4.1.11 Wet Scrubbing System

- Wet scrubbing system (Pilani is preferred) is required for minimizing expulsion of duct and coating resins from coating machine.
- Scrubber shall be supplied in MS with inlet & outlet connections fed through suitable piping by pump from water holding tank.
- Auto Dosing System for addition of Anti-Foaming agent should be provided.
- Make up water line & Drain line shall be provided with actuators and shall be operated through HMI/IPC.
- Blower shall be provided in the exhaust air unit on anti - vibration pads.

4.1.12 Loading / unloading of tablets

- Suitable loading device with valve for loading of cores from a closed IBC (1000 L coating machine).
- Suitable unloading device for unloading of coated tablets in a PE/ALU bag.

4.1.13 Control Panel

- SS 304 flame proof or non-flame proof HMI / IPC Panel shall be provided.
- HMI / IPC should have the following features.
- Auto / Manual operation selectable through HMI / IPC
- Individual sub system shall be operational in manual / maintenance mode.

4.2 Technical Requirement

1. IPC recipe should contain following steps with n phases
 - i. Charging
 - ii. Preheating
 - iii. Heating
 - iv. Spraying/Coating
 - v. Drying
 - vi. Cooling
 - vii. Discharging

4.3 Utilities Requirements (Electricity & Compressed Air)

- Chiller Water Inlet temperature: 12 Deg C Power Source: 415 VAC, 3ph, 50 HZ
- Compressed Air: min 5 Kg/cm2 pressure
- Steam Pressure: 3.5 kg/cm2
- Chiller Water Inlet temperature: up to 10 deg C
- Water connection: 3 bar pressure

4.4 Other Requirements

S. No	Description	Requirement